Journal of Engineering Research 14 (2026) 1039–1048

Contents lists available at ScienceDirect

Journal of Engineering Research

journal homepage: www.journals.elsevier.com/journal-of-engineering-research

Dispatch under pressure: An investigation into the cognitive load of Kuwait's emergency responders

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ARTICLE INFO

Keywords:
Cognitive workload
Emergency dispatchers
NASA Task Load Index (NASA-TLX)
Cognitive ergonomics
Emergency response systems

ABSTRACT

Emergency dispatchers play a pivotal role in ensuring efficient and effective emergency response systems. Reducing emergency dispatcher cognitive workload is crucial for enhancing response efficiency and accuracy, ultimately saving lives by ensuring rapid and precise coordination of critical resources. This study aims to assess the impact of age, work shift, and experience on the cognitive workload experienced by emergency dispatchers in Kuwait. A series of survey questions, structured around the dimensions of the NASA-TLX, were administered to gather quantitative data from 110 emergency dispatchers. In addition to quantitative analysis, qualitative data were obtained through interviews conducted with 11 dispatchers. Analysis of survey responses revealed elevated NASA-TLX scores among the participants, indicating significant cognitive workload. Results indicated that years of experience significantly influenced the cognitive workload experienced by dispatchers. By categorizing responses into the NASA-TLX dimensions, the study identified the most prevalent factors contributing to cognitive workload among emergency dispatchers. This research provides valuable insights into the cognitive demands faced by emergency dispatchers in Kuwait, highlighting the importance of experience in managing workload effectively. The findings contribute to the development of strategies aimed at optimizing the cognitive ergonomics of emergency dispatch operations, ultimately enhancing the efficiency and quality of emergency response systems.

Introduction

Emergency dispatchers serve as the critical nexus between individuals in distress and the resources required for timely and effective emergency response [23]. Their role demands swift decision-making, effective communication, and cognitive agility under pressure [26]. The cognitive workload experienced by emergency dispatchers is multifaceted, encompassing the processing of complex information, decision-making under uncertainty, and coordination of resources in high-stakes situations [25]. Understanding and mitigating the cognitive workload of dispatchers is paramount to optimizing emergency response systems and ultimately saving lives.

Cognitive workload refers to the mental effort required to perform a task, especially under complex or stressful conditions. When cognitive workload exceeds a person's capacity, performance can suffer — increasing the likelihood of error, delaying response times, and contributing to operator fatigue. The NASA Task Load Index (NASA-TLX) has become a widely accepted framework for assessing this workload across multiple domains [1]. Its six dimensions — Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration — offer a multidimensional view of how operators experience task-related demands [4].

Despite growing international attention to dispatcher workload (e.g., [28,37]), no empirical research has examined cognitive workload among emergency dispatchers in Kuwait. This gap is critical, given that Kuwait's dispatchers operate in a distinct environment that differs significantly from the settings most commonly studied. In Kuwait, all emergency calls — police, fire, and medical — are routed through a single national dispatch center. Unlike other nations that divide services across specialized call centers, Kuwaiti dispatchers triage and assign resources across all emergency domains. This centralized structure introduces elevated cognitive demands, particularly in the absence of sector-specific support systems [2].

Additional context-specific stressors compound the workload: dispatchers are often the only point of contact for callers who may speak limited Arabic or English, yet only one or two dispatchers per shift are

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https://doi.org/10.1016/j.jer.2025.06.004

Available online 28 June 2025

typically fluent in both languages. Kuwait also lacks a dedicated non-emergency number, leading to a consistently high volume of calls — many of which involve minor traffic issues or non-urgent complaints. Estimates suggest that nearly 40 % of these calls fall outside the scope of true emergencies, yet dispatchers are still required to resolve each within three minutes to avoid formal penalties [3].

Dispatchers are responsible not only for answering calls, but for making high-stakes decisions about which services to dispatch. Unlike other systems where resource allocation may be automated or sector-driven, dispatchers in Kuwait are accountable for both over- and under-response, which further intensifies decision-making pressure. Compounding this is the technological limitation that prevents handling multiple calls simultaneously — increasing the pressure to swiftly and accurately conclude each call. Organizational instability further contributes to workload. There is no standardized training program; new hires are trained informally through observation. High turnover rates result in weekly fluctuations in staffing levels due to resignations, vacations, or maternity leave — all of which affect performance consistency and increase burden on remaining staff [36]. Culturally, dispatchers must also manage emotionally intense and socially sensitive cases such as suicide attempts, domestic violence, and fatal accidents — all while adhering to cultural norms around family involvement and gendered communication. These unique sociocultural pressures, as noted by Ahmed & Alkhamis [2], reflect the broader challenges of emergency response in Gulf states, yet have not been systematically studied in relation to dispatcher workload.

Taken together, these geographic, structural, and cultural factors make Kuwait an important case for cognitive ergonomics research. This study directly addresses the gap in regional literature and contributes to the global understanding of dispatcher workload by highlighting how environmental context shapes both perceived and measured cognitive demands. By comprehensively assessing the cognitive workload of emergency dispatchers using the NASA-TLX and qualitative interviews, this research provides actionable insight into the specific stressors facing dispatchers in Kuwait. The study's findings can support targeted interventions, training design, and improved organizational practices to mitigate dispatcher strain and enhance emergency response effectiveness.

Related work

To establish the framework for this study, previous research was explored on (1) Impact of high cognitive workload (2) Prevalence of high cognitive workload amongst emergency dispatchers and (3) Assessment tools of cognitive workload. The main conclusions of the review of earlier research, which formed the basis of the current investigation, are presented in this section.

Impact of high cognitive workload

Cognitive workload, referring to the mental demands of a task, has been extensively studied across various environments and domains. In a study conducted by Malekpour et al. [32], cognitive workload was assessed during a driving simulator task using subjective ratings and physiological measures. The findings indicated that increased cognitive workload was associated with higher heart rate variability and pupil dilation, suggesting the potential of these measures as objective indicators of workload.

In the field of aviation, researchers have examined cognitive workload in relation to flight operations. In a study by Yazgan et al. [43] NASA-TLX was used to assess the workload of airline pilots during different phases of flight. The results highlighted that certain tasks, such as approach and landing, were associated with higher mental demands compared to others. This information is crucial for improving cockpit design, crew resource management, and training programs to effectively manage workload in aviation.

In the healthcare setting, cognitive workload has been investigated in surgical environments. A study by Zheng et al. [44] utilized the NASA-TLX to assess the workload experienced by surgeons during laparoscopic procedures. The research found that factors such as the complexity of the surgical task and equipment limitations contributed to higher workload levels. Understanding the cognitive demands in surgery can help in optimizing team coordination, procedure planning, and training methods to enhance patient safety and surgeon performance.

In the field of human-computer interaction, researchers have explored cognitive workload in relation to technology interfaces. A study by Devos et al. [10] examined the cognitive workload of users interacting with different smartphone applications using the SWAT scale. The results demonstrated that certain interface features, such as complex navigation and extensive content, increased cognitive workload. This research highlights the importance of designing intuitive and user-friendly interfaces to minimize cognitive demands and improve user experience.

In ergonomics, cognitive workload has been studied in relation to office work and information processing tasks. A study by Virtanen et al. [42] investigated the impacts of cognitive workload on office workers using subjective ratings and physiological measures. The findings revealed that high cognitive workload levels were associated with increased perceived stress and decreased task performance. This research emphasizes the importance of optimizing work environments, task design, and workload distribution to promote employee well-being and productivity.

Additionally, cognitive workload has been examined in the context of military operations. In a study by Dias et al. [11], cognitive workload was evaluated using subjective ratings and eye tracking technology during a military surveillance task. The findings highlighted the relationship between high cognitive workload and decreased visual attention to critical information. Understanding workload dynamics in military settings can assist in designing effective training programs, decision support systems, and equipment interfaces to enhance situational awareness and mission success.

These studies demonstrate the diverse applications of cognitive workload assessment in various environments. By measuring cognitive workload using subjective and objective measures, researchers can gain insights into the mental demands of different tasks and design interventions to optimize task performance, safety, and user experience. Whether in driving, aviation, healthcare, human-computer interaction, or other domains, understanding and effectively managing cognitive workload is crucial for ensuring optimal task performance and human well-being.

Prevalence of high cognitive workload amongst emergency dispatchers

Emergency dispatchers play a crucial role in ensuring effective emergency response. The cognitive workload experienced by these professionals is often high due to the nature of their work, which requires quick decision-making and multitasking [41]. Several studies have explored the cognitive workload of emergency dispatchers and its impact on performance and stress levels.

In a study by Tavares, Eva [40], the cognitive workload of emergency dispatchers was investigated using subjective ratings and physiological measures. The findings revealed that emergency dispatchers experienced significant mental demands, particularly during high-stress situations. The study also highlighted that increased cognitive workload was associated with higher levels of stress and fatigue. These findings emphasize the need for effective training programs and support systems to minimize the impact of cognitive workload on emergency dispatchers' well-being and performance.

Another study by Laguna et al. [29] examined the cognitive workload of emergency dispatchers during simulated emergency situations. The researchers used objective measures such as eye movement analysis and task performance metrics. The results demonstrated that emergency dispatchers experienced high levels of cognitive workload, which had implications for their ability to accurately process and prioritize information. This research underscores the importance of optimizing decision-support systems and providing appropriate resources to help emergency dispatchers manage their workload effectively.

Moreover, a study by Kosch et al. [28] explored the cognitive workload of emergency dispatchers when coordinating responses to multiple simultaneous incidents. The researchers observed that the cognitive workload increased significantly when dispatchers had to handle multiple incidents simultaneously. This heightened workload led to increased response times and potential errors in dispatching resources. The research suggests the need for improved task management strategies, workload assessment tools, and training protocols to enhance the performance of emergency dispatchers in high-demand situations.

Taken together, these studies highlight the demanding nature of the cognitive workload experienced by emergency dispatchers. Effective management of this workload is crucial for maintaining performance, minimizing stress, and ensuring efficient emergency response [27]. Strategies such as optimal training, supportive technologies, and workload assessment tools can contribute to improving the well-being and performance of emergency dispatchers, ultimately benefiting both emergency responders and the individuals in need during critical situations.

Cognitive workload assessment tools

NASA task load index (NASA-TLX)

NASA Task Load Index (NASA-TLX) is a well-established and widely used subjective workload assessment tool that provides valuable insights into the perceived cognitive workload experienced by individuals during various tasks [31]. NASA-TLX has found applications in various domains, including aviation, healthcare, human-computer interaction, and many others [22]. One of its primary advantages is its multidimensional nature, which allows for the assessment of different aspects of workload, including mental demand, physical demand, temporal demand, effort, performance, and frustration [20]. This comprehensive measurement approach makes NASA-TLX highly versatile and adaptable to different contexts.

In the field of aviation, NASA-TLX has been extensively utilized to evaluate pilots' workload in flight operations, simulator training, and human factors research [8]. By examining different workload dimensions, researchers and practitioners can identify areas that may cause excessive cognitive demands or affect the performance and safety of pilots. NASA-TLX has also been utilized in healthcare settings to assess the cognitive demands faced by healthcare professionals, such as surgeons, nurses, and anesthetists [6]. Understanding workload in healthcare can aid in improving task allocation, resource management, and patient safety [13].

In the domain of human-computer interaction, NASA-TLX has been employed to evaluate the cognitive demands associated with technology interfaces, software designs, and interactive systems [15]. This allows researchers and developers to identify potential areas where the interface design may be too mentally demanding, leading to user frustration or decreased performance. By using NASA-TLX, designers can make informed decisions when optimizing user interfaces to enhance usability, user experience, and overall task efficiency [15].

Subjective workload assessment technique (SWAT)

The Subjective Workload Assessment Technique (SWAT) is a self-report rating scale that has been widely used to assess cognitive workload in various domains. SWAT allows individuals to subjectively evaluate their workload experience using different dimensions such as mental demand, physical demand, temporal demand, performance, effort, and frustration [31]. This tool has found applications in areas such as aviation, driving, ergonomics, and human factors research.

In aviation, SWAT has been utilized to assess pilots' workload during flight operations, flight simulations, and cockpit designs [38]. By measuring different workload dimensions, SWAT assists in identifying situations where the cognitive demands on pilots might exceed their capacity, potentially leading to errors or reduced performance. This information is valuable for designing aircraft systems and procedures that effectively distribute workload and optimize pilots' performance, safety, and decision-making.

In the field of driving, SWAT has been applied to evaluate the cognitive workload experienced by drivers in various scenarios, such as different road conditions, traffic densities, and vehicle technologies [12]. By understanding the workload demands, researchers and policymakers can develop strategies to mitigate driver distraction, improve road safety, and design user-friendly in-vehicle technologies. For example, SWAT has been used to assess the workload impact of using hands-free mobile devices while driving [12].

Moreover, SWAT has found applications in ergonomics and human factors research to evaluate the cognitive demands associated with different work environments, tasks, and technology interfaces. By collecting subjective workload ratings from participants, researchers can analyze and optimize working conditions, workflows, and equipment design to enhance overall performance, reduce errors, and promote worker well-being and efficiency [39].

Research objectives

In light of prior work, the principal objective of this study is to investigate the factors affecting cognitive workload experienced by emergency dispatchers in Kuwait. The investigation is guided by two primary research questions:

1. What are the effects of work shift and professional experience on the cognitive workload of emergency dispatchers?
2. What factors may be influencing the cognitive workload experienced by emergency dispatchers?

By addressing these research questions, the study aims to provide insights that can help develop specific interventions and strategies to improve dispatcher performance and well-being and operational efficiency.

Methods

To achieve the research objectives, a mixed-methods approach was employed, combining a survey to measure NASA-TLX dimensions and qualitative interviews. Specifically, an electronic survey was distributed to 237 emergency dispatchers in Kuwait, covering the entire population of dispatchers during the study period. The survey was conducted in the autumn of 2023 and included dispatchers of both genders, regardless of their experience, work shift, or occupational level. However, individuals who had transitioned from emergency dispatch roles to managerial positions were excluded. A total of 110 responses were collected, following prior approval from the Ethical Review Board of Kuwait University for the research protocol.

To complement the quantitative data, 11 semi-structured interviews were conducted with randomly selected emergency dispatchers. These interviews were designed using the six dimensions of the NASA-TLX framework: frustration, performance, effort, physical demand, mental demand, and temporal demand. The interviews were transcribed and analyzed using thematic analysis to identify recurring patterns and themes. A codebook was developed to associate these themes with the NASA-TLX dimensions.

This section outlines the methods used for data collection, including the procedures for survey distribution and the interview process, as well as the tools employed for gathering and analyzing the data.

Procedure and data collection

The survey was distributed digitally through an official internal communication platform used by the national dispatch center. This platform is managed by shift supervisors and the center's chief officer, and membership is restricted to active emergency dispatchers. Only authorized personnel are permitted to send messages, ensuring that the survey link was delivered to all 237 active dispatchers during the study period. A total of 110 responses were received, representing dispatchers across all work shifts and levels of experience.

Prior to distribution, participants were provided with a clear explanation of the study's objectives, informed of its minimal risk nature, and assured of their right to withdraw at any time without penalty. The survey instrument, based on the NASA-TLX framework, was designed to assess various dimensions of cognitive workload experienced by emergency dispatchers. Participants submitted their responses through Microsoft Forms, the designated data collection platform. On average, completion time was approximately 4.6 min. Upon receipt, each submission was reviewed for completeness and accuracy to ensure the integrity of the dataset prior to analysis.

The interviewing phase of the project entailed conducting semi-structured interviews with 11 randomly selected emergency dispatchers from different work shifts. These interviews were conducted within the span of one working day. Each interview took place in a designated office setting, carefully selected to minimize distractions and facilitate privacy. Prior to commencing the interview, participants were provided with a comprehensive overview of the study's objectives and the purpose of the interview. Participants were then requested to provide consent for audio recording, with assurance that recordings would be used solely for research purposes and subsequently deleted upon project completion. Furthermore, participants were informed that any personal references to themselves would be redacted from the recordings. Subsequently, participants were prompted to respond to a series of questions, ranging from direct inquiries to open-ended prompts, aimed at exploring various dimensions of cognitive workload experienced by emergency dispatchers and their corresponding experiences. The duration of interviews varied, ranging from 8 to 24 min, reflecting differences in the level of detail provided by participants. Following the interviews, the researcher manually transcribed the recordings on a secure device to facilitate further analysis.

All procedures involving human participants were reviewed and approved by the Kuwait University Research Ethics Committee prior to data collection. Informed consent was obtained from all participants—both survey respondents and interviewees—through a written statement presented at the beginning of the survey and reiterated before each interview. The consent form outlined the study's purpose, the voluntary nature of participation, the right to withdraw at any time, and the measures taken to ensure confidentiality and data security.

To protect participant anonymity, no personally identifying information (e.g., names, ID numbers, or job titles) was collected. Survey responses were submitted anonymously through a secure online platform. Interview recordings were transcribed without personal identifiers, and all recordings were permanently deleted after transcription and verification.

Recognizing that participants were employed within a government-managed dispatch center, potential social desirability bias was considered. To minimize this risk, participants were assured that their responses would remain confidential, that no individual data would be shared with supervisors or administrative personnel, and that participation—or lack thereof—would carry no professional consequences. These procedures were implemented to promote honest disclosure and uphold ethical standards throughout the study.

Survey and interview instruments

The survey is composed of multiple sections, each of which fulfils a unique objective. The preliminary section consists of demographic inquiries designed to obtain data on the gender, age, years of experience, and work schedule of the respondents. The provision of descriptive demographic data enables the characterization of the sample population with respect to the distribution of work shifts, years of experience, and age distribution. Furthermore, it facilitates the ability to conduct comparative analyses of the total cognitive workload scores obtained from various work shifts and experience levels.

The survey then transitions to the cognitive workload assessment section, comprising a set of questions aligned with the dimensions of the NASA-TLX framework: Frustration, Performance, Effort, Physical, Mental, and Temporal Demand. Two questions were provided for each category, with there being 12 questions in total. These questions were adapted to suit the context of emergency dispatch operations. The English version of NASA-TLX was sourced directly from the original instrument [18], while an Arabic version was obtained from a previously validated study for verification purposes [17]. Responses were rated on a Likert scale ranging from 1 to 5, with 1 indicating "very low" and 5 indicating "very high" cognitive workload levels.

Similarly, the interview protocol is structured into distinct segments to capture comprehensive qualitative insights. The initial part focuses on gathering background information on interviewees, aiming to discern the influence of age, experience, and work shift on cognitive workload. Subsequently, the "A Day in the Life" segment entails open-ended inquiries concerning the daily routines and workplace dynamics of emergency dispatchers, elucidating the impact of environmental factors, training adequacy, and incident complexity on cognitive workload. Lastly, the third section delves into external distractions and unexpected changes, aiming to assess their effects on cognitive workload. Questions within these sections vary, ranging from open-ended inquiries to targeted yes-or-no inquiries, each designed to elicit specific aspects of cognitive workload experiences among emergency dispatchers.

Data analysis and results

To answer the first research question, statistical analyses were performed on the survey data using Minitab (version 18), and a significance level of 0.05 was used in all analyses. NASA-TLX scores were calculated and classified according to Galy (2018). To answer the second research question, interview data were transcribed and coded for thematic analysis. The results are presented in the following section.

Survey data from 110 emergency dispatchers (Table 1) indicate varying cognitive workload scores across different dimensions, ranging from "moderate" to "very high", based on Galy's scale (2018), with a mean aggregate score of 61.76 falling within the "very high" range. Effort scores the highest with a mean score of 13.94. A summary of the survey findings is included in Table 1.

To assess the internal consistency of the NASA-TLX instrument in this study, Cronbach's alpha was calculated across all six dimensions. The results indicated strong internal reliability, with an overall alpha of 0.85. This suggests that the instrument demonstrated a high degree of internal consistency when applied to the dispatcher population in Kuwait.

Each NASA-TLX dimension was analyzed using descriptive statistics, including 95% confidence intervals, to assess the precision of the

Table 1
NASA-TLX scores from survey.

Variable	Mean	SE Mean	StDev
Mental	11.91	0.23	2.42
Physical	10.64	0.34	3.56
Temporal	9.28	0.34	3.54
Effort	13.94	0.15	1.59
Performance	7.16	0.20	2.11
Frustration	8.84	0.33	3.46
Aggregate	61.76	1.16	12.21

estimated workload scores. The Effort dimension had a mean score of 13.94 with a 95% CI [13.64, 14.23], indicating consistently high perceived workload with relatively narrow variability. The Frustration dimension showed a mean of 8.84 with a 95% CI [8.19, 9.49], suggesting a moderately elevated emotional burden. Mental Demand had a mean score of 11.91 with a 95% CI [11.46, 12.37], and Temporal Demand had a mean of 9.28 with a 95% CI [8.61, 9.95], both reflecting the persistent time-sensitive and cognitive demands placed on dispatchers. The Performance dimension showed a mean of 7.16 with a 95% CI [6.76, 7.57], while Physical Demand had a mean score of 10.64 with a 95% CI [9.97, 11.28], slightly lower than cognitive dimensions but still notable. These intervals provide a clearer picture of the distribution and reliability of perceived workload scores across dimensions.

Q1. What are the effects of work shift and professional experience on the cognitive workload of emergency dispatchers?

This research question was developed to investigate how work shift and professional experience affect the cognitive workload of emergency dispatchers. It was hypothesized that dispatchers experience increased cognitive workload in the night shift [41] and that dispatchers with more experience have lower cognitive workload [29].

In order to assess the impact of work shift and years of experience on cognitive workload, seven two-way ANOVAs were computed with the independent variable being NASA-TLX dimension (mental, physical, temporal, effort, performance, and aggregate) and the dependent variables being work shift and years of experience. Prior to these analyses, assumptions were checked. Specifically, outlier analysis was conducted using Grubbs' test and results indicated no outliers for all variables ($p > 0.05$). Normality was confirmed for each dependent variable through the Anderson-Darling test ($AD < 1$, $p > 0.05$). Finally, Levene's test found no violations for the assumption of homogeneity of variances by cognitive workload scores ($p < 0.05$). All assumptions were met according to [7,34] therefore, analysis proceeded.

Experience had a statistically significant effect across all six NASA-TLX dimensions, with moderate effect sizes ranging from $\eta^2_p = 0.059$–0.089. The strongest effects were observed for Effort ($\eta^2_p = 0.089$, $p = 0.003$), Performance ($\eta^2_p = 0.081$, $p = 0.005$), and Temporal Demand ($\eta^2_p = 0.072$, $p = 0.008$), indicating that more experienced dispatchers consistently reported lower workload. Work shift alone did not have a significant effect on any dimension (all $p > 0.20$, $\eta^2_p < 0.02$). However, interaction effects between shift and experience were statistically significant for Effort ($\eta^2_p = 0.071$, $p = 0.009$), Mental Demand ($\eta^2_p = 0.078$, $p = 0.004$), and Temporal Demand ($\eta^2_p = 0.062$, $p = 0.016$), suggesting that workload is highest for inexperienced dispatchers working morning shifts. These findings support our original interpretation while providing clearer insight into the relative strength of each effect.

To understand how cognitive workload was impacted, Tukey's pairwise comparisons were run with 95 % confidence intervals. Post hoc tests revealed dispatchers with 6 years of experience had significantly lower cognitive workload than those with 0 years (Mean difference = -24.47, $p = 0.001$), while those with 17 years had lower workload than those with 1 year (Mean difference = -23.35, $p = 0.005$). However, there were no significant differences between those with 2 and 3 years ($p = 1.000$), or between those with 16 and 18 years ($p = 1.000$). These findings indicate that dispatchers experience a gradual reduction in cognitive workload after a significant number of years has passed, further investigation is required to determine the reason behind this.

Furthermore, the ANOVAs revealed statistically significant interaction effects between work shift and experience for aggregate scores ($F(40,70) = 4.0$, $p = 0.001$) and all cognitive workload dimensions (Mental ($F(38,88) = 1.45$, $p = 0.048$), Physical ($F(38,88) = 2.19$, $p = 0.012$), Temporal ($F(38,88) = 1.78$, $p = 0.036$), Effort ($F(38,88) = 1.65$, $p = 0.048$), Performance ($F(38,88) = 2.01$, $p = 0.019$), and Frustration ($F(38,88) = 1.58$, $p = 0.041$)) shown in Table 1, indicating that although work shifts alone may not significantly affect cognitive workload, the effect that experience has on cognitive workload depends on the shift the dispatcher works in. For instance, dispatchers with 17 years of experience showed significantly lower aggregate cognitive workload scores during the morning shift compared to the afternoon and night shifts. Similarly, dispatchers with 0 years of experience exhibited higher aggregate cognitive workload scores during the morning shift compared to the afternoon and night shifts (Fig. 1).

These results partially refute the initial hypothesis that night shift dispatchers experience higher cognitive workload than others. Specifically, results indicated that dispatchers consistently experience "very high" cognitive workload as per Galy (2018) regardless of their shift. However, experienced dispatchers in the morning shift exhibit lower high cognitive workload than their counterparts in other shifts, while inexperienced morning shift dispatchers exhibit higher cognitive workload than their counterparts in other shifts. Given the critical nature of calls during the morning shift [29] and the criticality of emergency dispatching operations overall [41], these findings call for inspection of factors contributing to the consistency of high cognitive workload amongst emergency dispatchers, with emphasis on those working in the morning shift. Further research is required to determine the factors that may be influencing cognitive workload experienced by emergency dispatchers.

Q2. What factors may be influencing the cognitive workload experienced by emergency dispatchers?

To investigate the factors influencing the cognitive workload experienced by emergency dispatchers, interviews were conducted with 11 randomly selected dispatchers across various work shifts and levels of experience. The purpose of this investigation was to understand and categorize the challenges faced by emergency dispatchers in their day-to-day operations.

Thematic analysis was conducted using an inductive coding approach guided by Braun and Clarke's (2006) framework. All interview transcripts were first reviewed in full to establish familiarity with the data. An initial codebook was then developed based on repeated patterns and emergent themes, with preliminary codes aligned to the six NASA-TLX workload dimensions. The data was coded at the sentence level. The data was coded by two raters (an Assistant Professor of Industrial Engineering, and an Industrial Engineering Graduate Student) and acceptable interrater reliability (Cohen's Kappa > 0.85) was observed [30]. To ensure consistency, coders met regularly to compare coding decisions and reconcile any differences. Discrepancies were resolved through iterative discussion and, when necessary, refinement of code definitions to improve clarity and alignment. Inter-coder agreement was assessed during the pilot phase using a subset of transcripts; agreement exceeded 85 %, which was deemed acceptable for the final analysis. When interpretive differences arose, decisions were guided by consensus and grounded in the participants' language and contextual cues within each transcript.

The finalized codebook offers a comprehensive description of each NASA-TLX dimension tailored to emergency dispatch tasks. Each dimension is supported by corresponding themes and descriptions, providing an in-depth understanding of the workload challenges reported by participants. The codebook was used to determine the frequency of each theme, with observations ranging from 0 to 15 across 96 coded statements. Analysis of frequency revealed two distinct groups: themes discussed 10 times or more and themes discussed less than 10 times.

Fig. 2 illustrates the percentage distribution of each theme based on all coded statements. Themes discussed 10 times or more included multitasking demands, emotional stress, task complexity, and time pressure. Multitasking demands accounted for ~20 % of total observations, making it the most frequently reported challenge. This theme aligns primarily with the Effort dimension, reflecting the mental

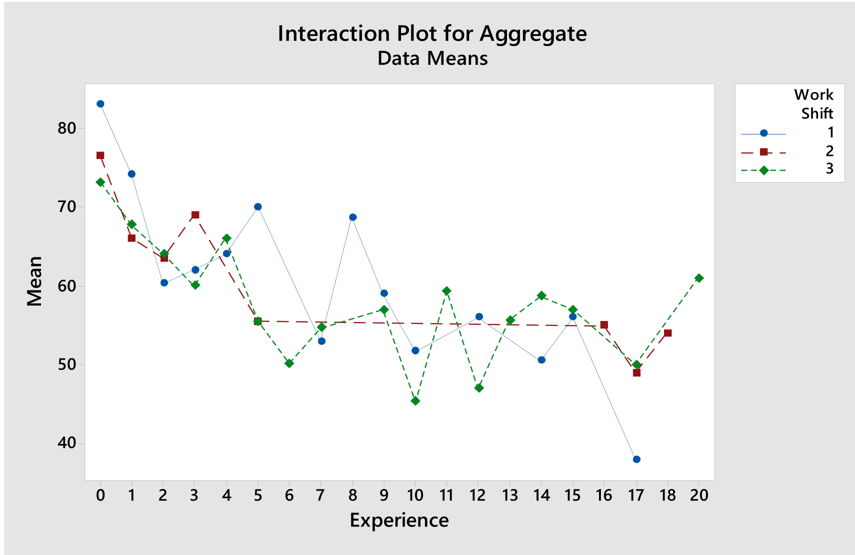

Fig. 1. Interaction Plot for Aggregate Scores.

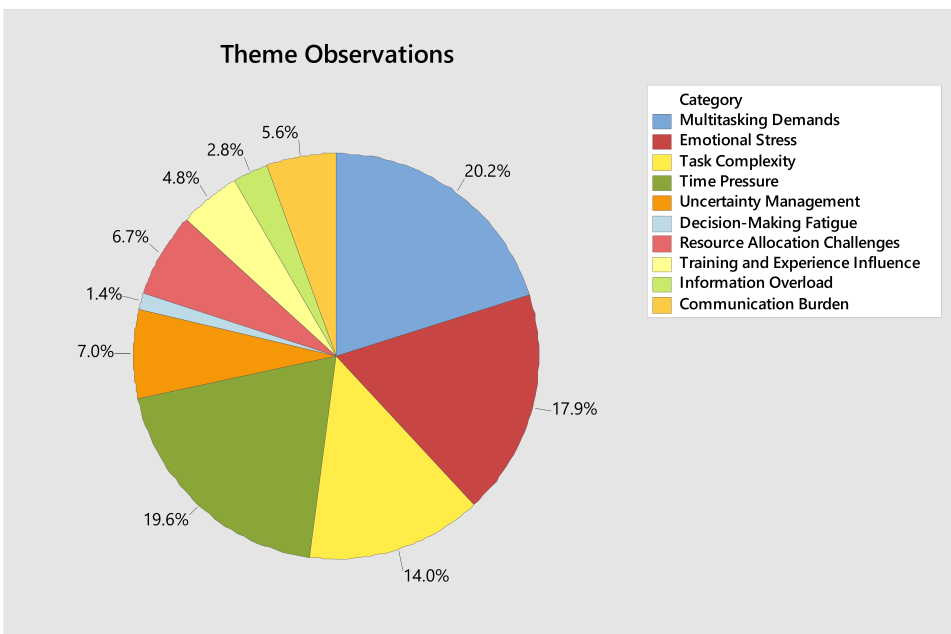

Fig. 2. Pie Chart of Theme Observations.

exertion required to handle simultaneous inputs and decisions. Emotional stress appeared in ~18 % of coded responses. It was associated with both the Effort and Frustration dimensions, underscoring the emotional toll dispatchers experience when handling life-threatening calls. Task complexity was cited in ~14 % of observations and related to Mental Demand, Effort, and Performance, particularly in situations such as locating callers without tracking systems. Time pressure was present in another ~20 % of observations and mapped strongly to the Temporal Demand dimension. It reflected urgency linked to the three-minute call handling rule, which adds pressure to collect information efficiently.

Less frequent but still relevant themes included uncertainty management, decision-making fatigue, resource allocation challenges, training and experience influence, information overload, communication burden, and personal life management. These occurred in 1–7 % of statements but often offered insight into context-specific stressors: Resource allocation challenges (~7 %) were linked to Mental Demand, Performance, and Effort, and reflected the strain of balancing calls when resources are scarce. Training and experience influence (~7 %) tied to Performance and Effort, and highlighted gaps in formal training and preparedness. Communication burden (~5.6 %) mapped to Effort and Frustration, and described issues with language barriers and poor audio quality. Decision-making fatigue (~1.4 %) was linked across four dimensions — Effort, Mental Demand, Performance, and Frustration — and referenced cognitive depletion during high-pressure judgment calls.

Fig. 3 presents the frequency of each NASA-TLX dimension across all coded statements. Because multiple dimensions could be associated with a single theme, some overlap was expected. The analysis revealed that Effort was the most frequently referenced dimension — appearing across all high-frequency themes. This finding reinforces the conclusion that Effort plays a central role in shaping dispatcher workload. Other dimensions — Frustration, Mental Demand, Performance, and Temporal

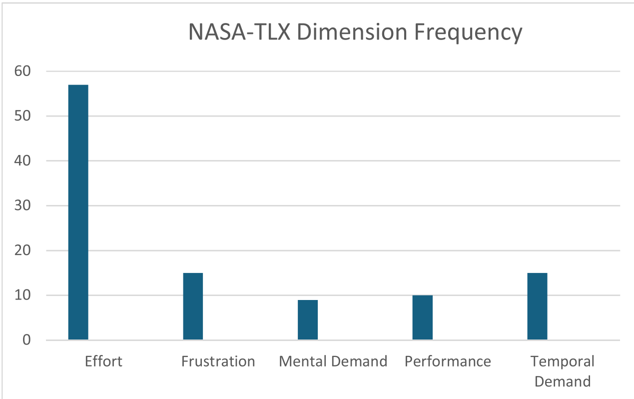

Fig. 3. NASA-TLX Dimension Frequency.

Demand — appeared at similar but lower frequencies, underscoring their contextual relevance depending on the nature of the task or challenge.

Discussion

The main goal of the study was to investigate the factors affecting cognitive workload experienced by emergency dispatchers in Kuwait. A survey was conducted to collect data from the dispatchers based on the NASA-TLX index. Interviews were also conducted to provide insight into reoccurring themes and factors affecting perceived cognitive workload. The data analysis was conducted using Minitab software program. The main findings of the study are as follows:

1. Emergency dispatchers in Kuwait experience high cognitive workload, especially in the Effort dimension, with variations in their workload based on experience and work shift.
2. Experience is negatively correlated with cognitive workload, and its effect varies depending on the dispatcher's shift. However, work shift alone does not significantly influence cognitive workload.
3. Several key themes influence cognitive workload including multitasking demands, emotional stress, task complexity and time pressure. All key themes influencing cognitive workload are directly related to the Effort dimension.

The results from this study can be used to contribute to a deeper understanding of the factors influencing the cognitive workload of emergency dispatchers and provide valuable insight for improving operational efficiency and dispatcher mental well-being.

Prevalence of high cognitive workload

While it is well-established in the global literature that emergency dispatchers face high cognitive workload, this study provides context-specific validation of that pattern within Kuwait — a dispatch environment that had not previously been studied. Rather than presenting this finding as novel in itself, the study emphasizes how it extends existing knowledge by identifying which specific dimensions of workload are most elevated in this setting (particularly Effort and Frustration), and by linking those findings to local operational and sociocultural stressors that are not typically examined in prior dispatcher research.

This study showed a prevalence of high cognitive workload scores among emergency dispatchers, with a mean cognitive workload score of 61.76, with Effort providing the highest sub-score, suggesting that emergency dispatchers experience substantial mental and physical effort in managing their tasks. In alignment with these findings, high cognitive workload scores were found to be prevalent amongst emergency dispatchers in other literature [29,33,41]. Terrell et al. [41] reported a mean cognitive workload score of 65.20 among emergency dispatchers in New York, however there is no mention of the highest sub-score category. Similarly, Marques et al. [33] reported similar findings in healthcare emergency dispatchers with high cognitive workload, although the exact mean is not reported. Moreover, Laguna et al. [29] validated these findings with a reported mean cognitive workload score of 70.5 amongst emergency dispatchers, further corroborating the prevalence of high cognitive workload in this profession. This could be due to work environment factors such as noise, interruptions and light levels, since researchers have found that noise and light levels tend to have a great impact on high cognitive workload. Acosta [1] Kanaan [24]; Marques et al. [33] also suggests that lack of autonomy in the dispatcher profession could also contribute to the prevalence of high cognitive workload amongst dispatchers. Mohammadian [35] however suggests the opposite; that high cognitive workload may be the result of too much workplace autonomy, contributing to decision-making fatigue, in the context of control room operations. Further research is required to understand the prevalence of high cognitive workload amongst dispatchers, while accounting for other workplace factors such as but not limited to equipment used, training methods, and work schedules.

Impact of experience and work shift on cognitive workload

The study investigated the impact of experience and work shift on cognitive workload scores, as guided by previous research. The emphasis was placed on identifying the relationship between years of experience and cognitive workload as well as type of work shift and cognitive workload. The results showed a significant negative relationship between professional experience and cognitive workload among emergency dispatchers, depending on the work shift of dispatchers. This aligns with the findings of previous papers that observed higher cognitive workload amongst unseasoned emergency dispatchers [15,16,21]. For example, Ghalenoiej et al. [15] found that those with less than one year of experience had significantly higher cognitive workload scores compared to those with more than five years of experience. Similarly, Grier [16] reported that dispatchers with less than two years of experience had significantly higher cognitive workload scores compared to those with more than 10 years of experience. Similarly, [21]) found a significant negative correlation between years of experience and cognitive workload scores ($r = -0.45$, $p = 0.012$). However, contrary to the findings of previous literature, the study conducted found no significant impact of work shift on cognitive workload among emergency dispatchers. There is, however, a significant interaction observed in how dispatchers' experience affects their cognitive workload depending on the shift they work in. This is specifically observed in the morning shift, in which inexperienced dispatchers had higher cognitive workload than their counterparts in other shifts. This may be primarily due to the fast-paced nature of calls received in the morning shift due to traffic. These finding contrast with prior research by Terrell et al. [41], who reported that dispatchers in the night shift experience the highest mean cognitive workload scores compared to their counterparts in other shifts, stating reasons such as severity of calls received and emotional stress. Other studies did not comment on the effects of work shift on mean cognitive workload scores [15,16,21]. These findings highlight the importance of optimizing shift scheduling practices to lower overall cognitive workload [29]. Organizations may wish to explore options such as fatigue management protocols to alleviate and mitigate cognitive workload associated with demanding work schedules [27].

Influential themes on cognitive workload

Thematic analysis of this study identified several themes influencing cognitive workload, including multitasking demands, emotional stress, task complexity, and time pressure. These themes were most frequently linked to the Effort dimension of the NASA-TLX, which had the highest number of coded references across interview statements. The prevalence of these themes, combined with the consistently elevated Effort scores observed in the quantitative data (mean = 13.94, 95 % CI [13.42, 14.46]), reinforces the conclusion that dispatchers experience high cognitive and emotional strain across shifts.

This convergence of data sources highlights how specific operational demands drive sustained mental effort. For example, multitasking demands — the most frequently mentioned theme — align with the elevated Effort scores and reflect the need to manage simultaneous calls, data entry, and triage decisions under time pressure. Emotional stress was also highly referenced and maps to both Effort and Frustration (mean = 12.86, 95 % CI [12.21, 13.51]), supporting the conclusion that dispatchers face an emotionally charged work environment that extends beyond cognitive complexity. Task complexity and time pressure further reinforce the quantitative results related to Mental and Temporal Demand dimensions, especially in the context of the three-minute call resolution policy.

Less frequently observed themes such as training and experience influence, communication burden, and decision-making fatigue were also aligned with NASA-TLX dimensions — particularly Performance and Frustration — but were noted in fewer statements. These lower-frequency themes nonetheless offer insight into systemic gaps, such as lack of formal training, difficulty communicating with non-Arabic-speaking callers, and cognitive fatigue during complex triage situations.

This integrated interpretation strengthens the validity of the findings and provides a foundation for actionable interventions. Several practical implications emerge. First, the strong association between high Effort scores and multitasking suggests a need to simplify dispatcher workflows. This could involve introducing real-time triage tools, structured decision aids, or automated call logging to reduce cognitive load. Second, training programs should incorporate simulation-based scenarios that mirror high-stress situations, particularly those involving ambiguous information or language barriers, to better prepare new dispatchers for real-world demands.

Given the interaction between shift and experience observed in the quantitative analysis, newer dispatchers may benefit from mentorship-based scheduling or assignment to lower-demand shifts during the early phase of employment. Finally, ergonomic design principles — such as reducing unnecessary information load, minimizing task switching, and improving system interface usability — could further reduce perceived workload and improve dispatcher well-being.

These integrated insights align with and extend prior research. For instance, Terrell et al. [41] highlighted the emotional strain of night-shift work, while this study identifies morning shifts as more cognitively taxing for less experienced dispatchers — pointing to the importance of local context. While previous studies such as Kosch et al. [28] categorized themes differently or emphasized alternate NASA-TLX dimensions (e.g., Performance, Mental Demand), the present findings support the dominant role of Effort and the need for complementary qualitative approaches in high-context dispatch environments like Kuwait.

Conclusion, limitations, and future work

This study investigated the cognitive workload experienced by emergency dispatchers in Kuwait and the factors influencing it. A survey based on the NASA-TLX framework was used to measure aggregate workload scores and sub-dimensions across six workload categories. To complement the quantitative data, qualitative interviews were conducted to explore dispatcher experiences and contextual stressors. Two-way ANOVAs were used to assess the effects of work shift and years of experience on both overall workload and each NASA-TLX sub-dimension. Results indicated that years of experience significantly influenced aggregate workload and all six sub-scores, while work shift alone had no significant main effect. Additional one-way ANOVAs and Tukey's post hoc tests revealed that less experienced dispatchers consistently reported higher levels of cognitive workload. Thematic analysis of interview data reinforced these findings. Frequently reported stressors, such as multitasking demands, emotional stress, task complexity, and time pressure, were most closely associated with the Effort dimension, which also received the highest mean score in the survey data. The integration of both data sources helped build a more comprehensive understanding of the demands placed on dispatchers in Kuwait's centralized emergency response system.

While the findings support the study's objectives, several limitations should be acknowledged. First, the sample size of 110 dispatchers, though diverse in shift and experience level, may limit the generalizability of results. Second, reliance on self-report measures and retrospective recall in both the survey and interviews introduces the possibility of subjective bias. Third, the cross-sectional design restricts the ability to draw causal inferences. Finally, as the study was conducted exclusively within Kuwait, findings may not be directly transferable to emergency dispatch systems in other cultural or organizational contexts [29].

The methodological design relied primarily on self-reported data obtained through structured surveys and semi-structured interviews. While this approach enabled detailed insight into dispatcher perceptions and experiences, it also introduces inherent risks of subjectivity and potential bias. The NASA-TLX framework was selected for its established validity in workload assessment [16,18], and its use in conjunction with qualitative interviews provided a mixed-methods structure that allowed for internal triangulation. Nevertheless, the absence of physiological workload measures, such as heart rate variability, galvanic skin response, or eye tracking, limits the ability to validate self-reported workload levels through objective metrics [10,19]. These tools were excluded due to both operational constraints and ethical considerations, particularly given the sensitive and fast-paced nature of emergency dispatch environments in Kuwait, where continuous physiological monitoring may disrupt workflow or violate participant privacy.

In addition, several potentially relevant variables were not controlled for, including dispatcher education level, type of emergencies handled, and broader organizational structure. However, in the context of Kuwait's emergency response system, some of these variables are inherently uniform. All emergency calls, across police, fire, and medical domains, are routed through a single, centralized national dispatch center [36]. This design standardizes operational procedures across all dispatchers and effectively removes variability in dispatch center size or emergency type. Dispatcher education level was not recorded due to institutional limitations and the absence of formal academic requirements for employment in these roles. As a result, the study focused on professional factors such as years of experience and work shift, which have been consistently associated with cognitive workload in prior research [11,24].

Although more advanced multivariate techniques such as MANOVA or hierarchical regression could provide additional insight, particularly when modeling the interdependence of variables or controlling for covariates, the decision to employ two-way ANOVAs was based on both conceptual and methodological considerations. Each NASA-TLX dimension represents a distinct aspect of cognitive workload (e.g., Mental Demand, Effort, Frustration), and these dimensions have been widely treated in the literature as separate constructs [9]. Accordingly, separate two-way ANOVAs were used to assess the individual and interaction effects of work shift and experience on each workload dimension, facilitating clearer interpretation of where significant effects and interactions occurred. To enhance the robustness of these analyses, effect sizes (partial eta squared) and confidence intervals were reported alongside p-values, providing information about both the strength and precision of observed effects. Furthermore, the interpretation of quantitative findings was supported through triangulation with thematically analyzed qualitative data, offering complementary contextual depth. Nonetheless, the use of more sophisticated statistical models in future studies, such as MANOVA or hierarchical regression, would enable the inclusion of additional covariates (e.g., age, gender, education level) and allow for a more comprehensive understanding of interaction effects and cross-dimensional relationships [37].

Although the NASA-TLX has been widely used in prior studies of dispatcher workload [11,38], this study contributes original value in three key areas: contextual, methodological, and practical. Contextually, it is the first to examine dispatcher workload within Kuwait's centralized and multilingual emergency dispatch system, which differs substantially from models in North America and Europe. Unique operational features, including high call volumes due to the absence of a non-emergency line, inconsistent training standards, limited bilingual coverage, and sociocultural expectations around emotional expression and family involvement, create distinct stressors that have not been previously documented in the dispatcher workload literature.

Methodologically, the study integrates a mixed-methods approach, combining quantitative NASA-TLX scores with qualitative thematic analysis to contextualize the experience of workload. This enabled alignment between workload dimensions and system-specific themes, such as emotional labor, multitasking strain, and time-pressured decision-making. This dual approach yields a deeper understanding of dispatcher demands than studies using quantitative instruments alone.

Substantively, the findings generate actionable recommendations tailored to Kuwait's dispatch environment. These include the implementation of structured dispatcher training, the introduction of real-time triage aids, and the adoption of mentorship-based scheduling practices for newly hired staff. These targeted interventions address locally specific challenges and offer a replicable model for examining dispatcher workload in other under-researched or culturally distinct contexts.

The results confirm that high cognitive workload is a sustained characteristic of emergency dispatch work, rather than a situational or episodic condition. Dimensions such as Effort, Frustration, and Temporal Demand consistently scored high, reflecting the inherent complexity, urgency, and multitasking required by the role [14,5]. These pressures are further intensified by systemic constraints, such as limited support tools and inconsistent training. Although professional experience may moderate some aspects of perceived workload, the findings underscore the need for structural interventions rather than reliance on individual adaptation. The implications extend beyond individual performance. Emergency dispatchers serve as the first point of contact in high-stakes emergencies, often making rapid decisions with limited information. Preserving the reliability of this function requires sustained institutional investment in workload mitigation, resource allocation, and operational reform. Addressing these systemic demands is essential not only to protect dispatcher well-being, but also to ensure the ongoing effectiveness of public emergency response systems

We would like to express our sincere gratitude to the General Department of Operations at the Ministry of Interior in Kuwait for their invaluable support in data collection. Their cooperation and assistance were instrumental in the successful completion of this study.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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